

1 Quick refresher on the current in electrodynamics

1.1 Maxwell's equations

Given Maxwell's equations,

$$\vec{\nabla} \cdot \vec{E}(\vec{r}, t) = \frac{1}{\varepsilon_0} \rho(\vec{r}, t) \quad (1)$$

$$\vec{\nabla} \cdot \vec{B}(\vec{r}, t) = 0 \quad (2)$$

$$\vec{\nabla} \times \vec{E}(\vec{r}, t) = -\frac{\partial}{\partial t} \vec{B}(\vec{r}, t) \quad (3)$$

$$\vec{\nabla} \times \vec{B}(\vec{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \vec{E}(\vec{r}, t) + \frac{1}{\varepsilon_0 c^2} \vec{j}(\vec{r}, t) \quad (4)$$

1.2 Continuity equation

From $\varepsilon_0 \partial_t$ (eq. (1)), using $\partial_x \partial_t = \partial_t \partial_x$,

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) = \varepsilon_0 \vec{\nabla} \cdot (\partial_t \vec{E}(\vec{r}, t)), \quad (5)$$

and from $\varepsilon_0 c^2 \vec{\nabla} \cdot$ (eq. (4)),

$$\begin{aligned} \vec{\nabla} \cdot \vec{j}(\vec{r}, t) &= \vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}(\vec{r}, t)) - \varepsilon_0 \vec{\nabla} \cdot (\partial_t \vec{E}(\vec{r}, t)) \\ &= -\varepsilon_0 \vec{\nabla} \cdot (\partial_t \vec{E}(\vec{r}, t)) \quad (\text{div}(\text{rot } \vec{F}) = 0). \end{aligned} \quad (6)$$

Adding the two gives the continuity equation

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) + \vec{\nabla} \cdot \vec{j}(\vec{r}, t) = 0. \quad (7)$$

1.3 Point-particle solutions

Solutions of the continuity equation:

$$\rho(\vec{r}, t) = \sum_{\alpha} q_{\alpha} \delta(\vec{r} - \vec{r}_{\alpha}(t)) \quad (8)$$

$$\vec{j}(\vec{r}, t) = \sum_{\alpha} q_{\alpha} \vec{v}_{\alpha}(t) \delta(\vec{r} - \vec{r}_{\alpha}(t)) \quad (9)$$

Proof. Note that

$$\begin{aligned} \partial_t \delta(\vec{r} - \vec{r}_{\alpha}(t)) &= \partial_t \left(\delta(r_x - r_x^{\alpha}(t)) \delta(r_y - r_y^{\alpha}(t)) \delta(r_z - r_z^{\alpha}(t)) \right) \\ &= - \sum_{i=x,y,z} \dot{r}_i^{\alpha}(t) \partial_{r_i} \delta(r_i - r_i^{\alpha}(t)) = -\dot{\vec{r}}_{\alpha}(t) \cdot \vec{\nabla} \delta(\vec{r} - \vec{r}_{\alpha}(t)), \end{aligned} \quad (10)$$

where we used the chain rule in the second line and

$$\begin{aligned}\vec{\nabla} \cdot \left(\vec{v}_\alpha(t) \delta(\vec{r} - \vec{r}_\alpha(t)) \right) &= \sum_{i=x,y,z} \partial_{r_i} \left(v_i^\alpha(t) \delta(\vec{r} - \vec{r}_\alpha(t)) \right) \\ &= \sum_{i=x,y,z} v_i^\alpha(t) \partial_{r_i} \delta(\vec{r} - \vec{r}_\alpha(t)) = \vec{v}_\alpha(t) \cdot \vec{\nabla} \delta(\vec{r} - \vec{r}_\alpha(t)).\end{aligned}\quad (11)$$

Therefore

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) = - \sum_{\alpha} q_{\alpha} \dot{\vec{r}}_{\alpha}(t) \cdot \left(\vec{\nabla} \delta(\vec{r} - \vec{r}_{\alpha}(t)) \right) = - \vec{\nabla} \cdot \vec{j}(\vec{r}, t), \quad (12)$$

which is (7). □

Note that these solutions are not unique (one equation for four components).

Intermediate result: the current can be defined by “charge times velocity”.